fox2db — controller model

Sofar / Waveshare ESP32-S3 6CH

Load-following PV-surplus controller for a solar plant with two battery stores. It switches a *quantized* electrical load (EBox) so that the grid exchange stays near zero (“sweet spot”), without disturbing the autonomous Sofar battery.

Classification

Discrete-time, nonlinear **hybrid automaton** (Mealy machine with hysteresis) — **not a PID, not an LTI system**. The actuator is discrete (8 power levels via 3 relay bits), hence dead-band + rate limiting instead of a continuous control signal. Cycle: 60 s.

Mathematical model

Power map and (power-sorted) rank:

$$P = (0,\ 3000,\ 3650,\ 6650,\ 3900,\ 7100,\ 7800,\ 11400)\ \mathrm{W}$$

$$\Pi_{\text{asc.}} = (0, 3000, 3650, 3900, 6650, 7100, 7800, 11400)$$

Step recursion, state $x_k = (s, c, E)$:

$$e_{\text{eff}} = \max\bigl(e_k,\ P_{s_{k-1}}\bigr) \quad (s_{k-1} > 0), \qquad b_k^{\text{eff}} = \begin{cases} b_k & b_k < 0 \text{ (Sofar discharging, full)} \\ \varphi\, b_k & b_k \geq 0 \text{ (Sofar charging, fractional)} \end{cases}$$

$$E_k = p_k + e_{\text{eff}} + b_k^{\text{eff}}$$

$$\tau_k = \begin{cases} 0 & E_k < E_{\min} \\ Q(E_k + G) & \text{otherwise} \end{cases} \qquad Q(B) = \max\{\, s : P_s \leq B \,\}$$

Up-ramp (max. one power step per cycle):

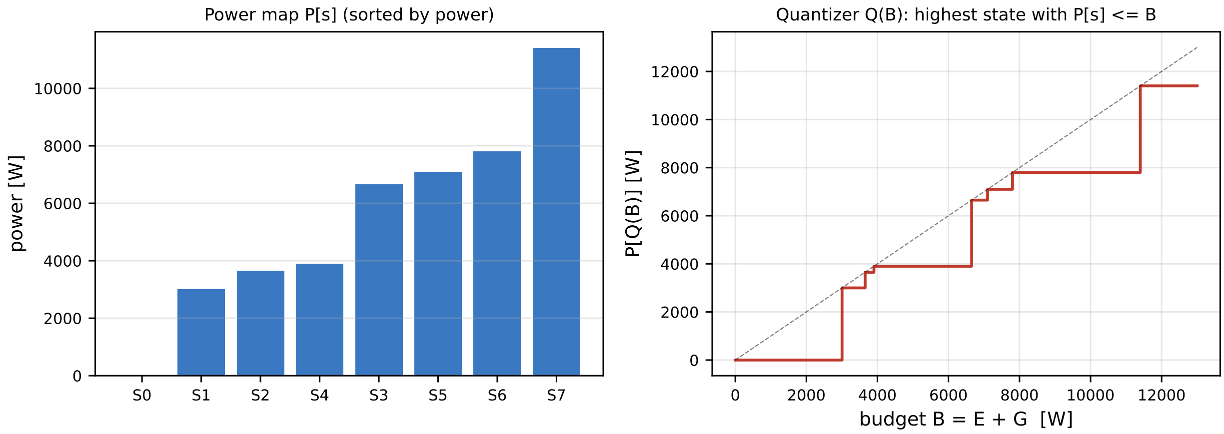
$$\tilde{\tau}_k = \min_{\text{rank}} \bigl(\tau_k,\ \hat{s}(\text{rank}(s_{k-1}) + 1)\bigr)$$

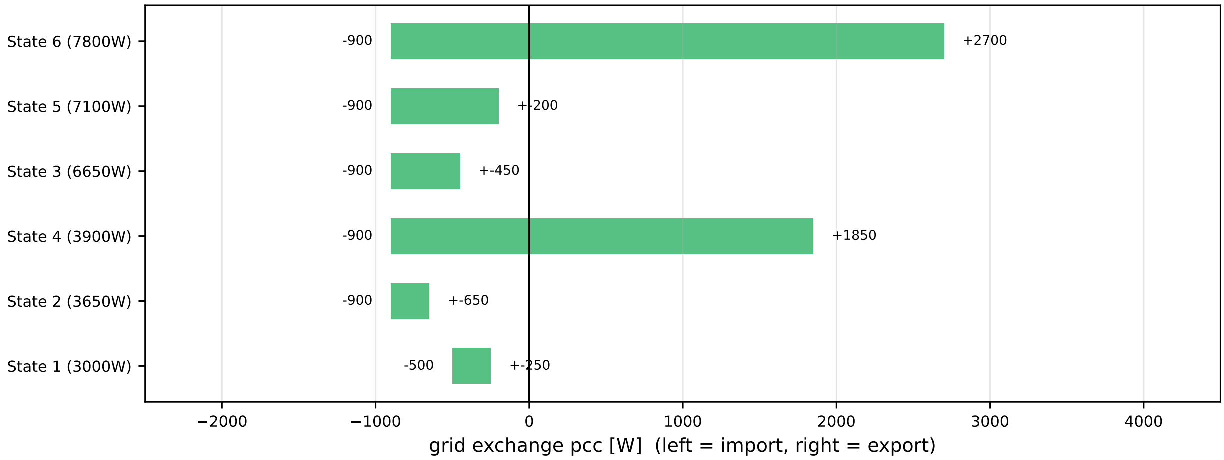
Switching / protection logic (order = priority):

Rule	Condition	Action
`EMERGENCY_FORCE`	pcc < -1020 ($= -(G{+}120)$) and down	switch now
`SWEET_SPOT_HOLD`	up and $\lvert\text{pcc}\rvert < 160$	hold
`TREND_BLOCK`	up and $\dot{E} < -20\,\mathrm{W/s}$	hold
`BAT_GUARD_BLOCK`	up and $b_1 < -110\,\mathrm{W}$	hold
`STABILIZING`	down and $c < 2$ cycles	hold
`HYSTERESIS`	down and $\lvert\Delta P\rvert < 505\,\mathrm{W}$	hold

Constants: $E_{\min} = 2500$, $G = 900$, $H = 505$, $N = 2$, $\text{EMERG} = G + 120$ [W]; $\varphi =$ `bat1_factor` (share of Sofar charging counted as surplus, default 0.5, MQTT `sofar/bat1_factor`).
Charge-block latch (hysteresis against chatter at the ratio threshold r_{th}): LOCK at $r_{\text{ist}} \leq r_{\text{th}}$, RELEASE only at $r_{\text{ist}} \geq r_{\text{th}} + 0.15$.
Dead-band (equilibrium, load on, $e_{\text{eff}} = P_s$):

$$\max(-G,\ E_{\min} - P_s)\ <\ p\ <\ \bigl(\Pi^{+}(s) - P_s\bigr) - G$$





Dead-band per state: lower edge $= \max(-G, E_{\min} - P_s)$, upper edge $=$ next power level $-P_s - G$.

Simulation

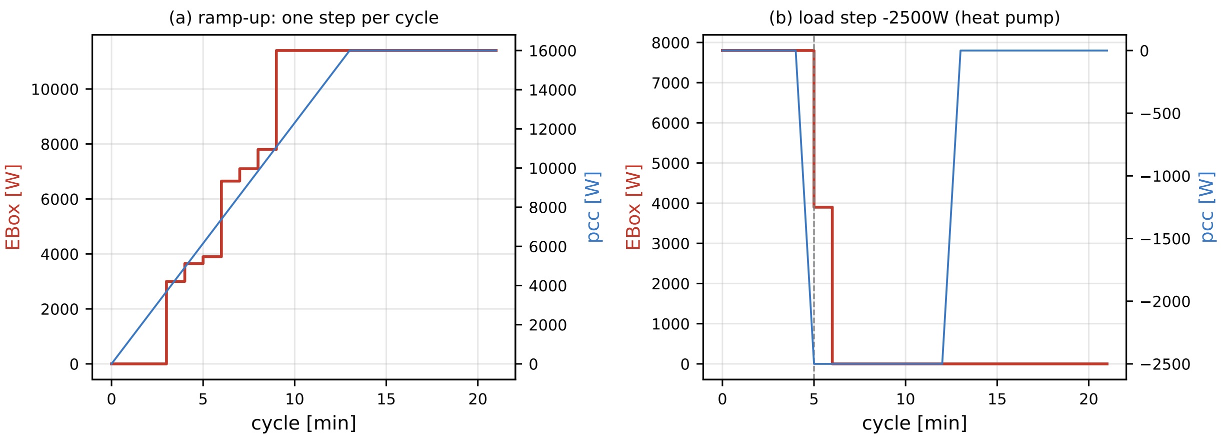
(a) Ramp-up: rate limiting allows only one power step per cycle. (b) Load step in the sweet spot: -2500 W from cycle 5 (heat pump) is shed within a few cycles.

Plant limits

Fuse 3×50 A @ 230 V $=$ 34.5 kW; PV peak 35 kWp; Bat1 (Sofar) 5 kWh; Bat2 (EBox) 30 kWh. EBox max (state 7) 11.4 kW $=$ 33 % of the fuse. Bat2 @ state 7: $+0.63$ %/min ($\rightarrow$ full in $\sim$2.6 h).

Validation

`step_full_literal` was checked one-step-ahead against the ESP's actual decisions (`pv_decision_log`): **55/55 state match (100 %)**, excess formula 54/55. Note: the C code compares by state *number* for the up-ramp rather than the power *rank*; since P is not monotone in the number (state4 $= 3900 <$ state3 $= 6650$ W), this deviates in edge cases from the rank-monotone model. `step_full_literal` mirrors the C code 1:1.



(a) ramp-up: one step per cycle
EBox [W]
pcc [W]
cycle [min]
(b) load step -2500W (heat pump)
EBox [W]
pcc [W]
cycle [min]